

AI Content Pipeline POC — Scope Document

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1. Overview

Embibe's content gap is heavily skewed toward non-STEM subjects, primary grades, and vernacular mediums — segments where manual production is slow, and hard to prioritize.

This POC runs four controlled experiments to test two AI-native production approaches side-by-side: one that curates existing open-source content, and one that generates original content from scratch.

The goal is to answer three questions:

1. **Is the output quality good enough to publish?**
 2. **Does it actually fill the content gaps Embibe has today?**
 3. **Is it faster and more cost-effective than how Embibe produces content today?**
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2. POC Objectives

This POC tests two production approaches:

- **Curation** — AI shortlists and packages existing open-source content (NCERT PDFs, vetted YouTube channels) against a specific chapter-topic combination.

- **Generation** — AI produces original Embibe-branded content end-to-end (script → video → voice → metadata).

The three objectives:

1. **Use Curation to cover the gaps we choose not to prioritize now** — primary grades, vernacular, and non-STEM subjects where manual production isn't commercially justified.
2. **Use Generation to speed up studio-quality production** — cut weeks of scripting and shooting/editing down to days, for core CBSE middle-school subjects where we want full brand and IP control.
3. **Validate every output against a single quality scorecard** — every topic, every approach, scored the same way, so we have measurable evidence (not opinion) on where each approach works and where it doesn't.

3. Experiments at a Glance

All experiments: CBSE board, 1 chapter × 3 topics each.

#	Approach	Gr	Subject	Chapter — Topics	Lang
1	Curation	5	English	The Rainbow (Santoor) — <i>Theme & meaning; New vocabulary; Poetic devices (rhyme, imagery)</i>	En
2	Curation	6	Hindi	Malahar — <i>Theme & imagery; Vocabulary & phrases; Poetic devices</i>	Hi
3	Gen + Curation	6	SST (History)	Timeline and Sources of History — <i>Dating conventions (BCE/CE, timelines); Sources of history (literary, archaeological, oral); How historians construct the past</i>	En
4	Gen + Curation	8	SST (Geography)	Natural Resources and Their Use — <i>Types of resources (renewable vs non-renewable, biotic vs abiotic); Distribution and human use; Conservation and sustainable use</i>	En

For Experiments 3 and 4, we run the **same topics through both Curation and Generation** — so we can compare the two approaches side-by-side on identical input and see which one produces better output for the same chapter.

4. Input

Every experiment starts with the same input, regardless of approach.

Field	Description
Board	CBSE
Grade	5 / 6 / 8
Subject	As per experiment
Chapter + Topics	1 chapter, 3 topics scoped per experiment
Language	English or Hindi
Learning objectives	4–6 per topic, sourced from NCERT syllabus
Reference material	NCERT textbook PDF for the chapter, NCFSE document, Youtube Channels

5. Output

5.1 Curation approach (Experiments 1, 2, and the curation pass of 3 & 4)

Artifact	Detail
Curated resource	1 YouTube video segment (with start/end timestamps) or 1 NCERT PDF excerpt (page range)
Topic summary	~300-word explainer in target language
Key concepts	5–7 tagged terms with definitions, flowcharts and infographics (as relevant)
Assessment	10 MCQs + 5 short-answer questions with model answers
Metadata	Learning objectives covered, prerequisites, difficulty, Bloom’s level
Source record	Original URL, creator, license type, attribution text

YouTube source channels (starting allowlist — curation is restricted to vetted educational channels): - Magnet Brains - LearnoHub — Class 6, 7, 8 - Next Toppers Juniors - NCERT Official - BodhaGuru Learning

5.2 Generation approach (Experiments 3 and 4)

Artifact	Detail
Explainer video	Original, 8–12 min, AI cloned voice, 1080p, captions embedded
Script	LO-aligned lesson with explanation, examples, check-for-understanding
Visual assets	Slides, diagrams, images; 1 simulation or 3D asset where topic warrants
Transcript + key moments	Timestamped transcript, 3–6 key moments for jump-to navigation
Assessment	10 MCQs + 5 short-answer questions with model answers
Metadata	Learning objectives covered, prerequisites, difficulty, Bloom’s level
Provenance record	Models used, voice-clone ID, generation timestamp

6. Tools & Models by Pipeline Stage

This section maps each stage of the pipeline to the purpose it serves and the specific tools and models we propose to use. Choices will be validated in Phase 1 of the execution plan; alternates are listed where a head-to-head bake-off is worthwhile.

6.1 Curation pipeline

Stage	Purpose / outcome	Proposed tools & models	Alternates
Source discovery (YouTube)	Find candidate videos on allowlisted channels for a chapter + topic	YouTube Data API v3; yt-dlp for download	—
NCERT PDF ingestion	Extract text, tables, figures, page ranges from textbook PDFs	PyMuPDF / pdfplumber for base extraction; Gemini 2.5 Pro (vision) for layout-aware parsing of diagrams and page spreads	LlamaParse, Unstructured.io, Mistral OCR
Transcript (if no captions)	Timestamped transcript from video audio	Whisper v3 Turbo (English); AI4Bharat IndicConformer or Sarvam ASR (Hindi/Indic)	Gemini 2.5 Flash audio, AssemblyAI

Stage	Purpose / outcome	Proposed tools & models	Alternates
Curriculum alignment	Match source content to the target chapter-topic-LO node in Embibe's curriculum graph	Existing Superr curriculum graph + embedding retrieval	—
Topic summary (~300 words)	Write a short, grade-appropriate explainer in the target language	Gemini 2.5 Pro (long context, strong Indic); Claude Sonnet 4.6 for English	GPT-5
Key concepts + definitions	Extract 5–7 tagged terms from source material with grade-appropriate definitions	Gemini 2.5 Pro / Claude Sonnet 4.6 with structured output	—
Flowcharts / infographics	Turn explanations into visual aids	Mermaid / Graphviz (LLM-generated source); Napkin AI or Excalidraw AI for richer visuals	Whimsical AI
Assessment (10 MCQs + 5 short-answer)	LO-aligned questions with model answers	Gemini 2.5 Pro / Claude Sonnet 4.6 with Bloom's-level prompting	—
Metadata + source record	LO coverage, Bloom's, difficulty, licence, attribution	LLM tagging + deterministic licence lookup (Creative Commons API, YouTube API)	—

6.2 Generation pipeline

Stage	Purpose / outcome	Proposed tools & models	Alternates
Scripting	Write an 8–12 min LO-aligned lesson: explanation, examples, check-for-understanding, recap	Gemini 2.5 Pro (long context, Indic-strong); Claude Opus 4.7 for pedagogical tone; GPT-5	—
Slide / deck generation	Lesson slides aligned to the script	Gamma (text-to-deck) or LLM → Marp / Reveal.js (programmatic, versionable)	Beautiful.ai, Tome
Diagrams	Concept diagrams, labelled figures, timelines	Excalidraw AI , Napkin AI , Mermaid	Whimsical AI
Image generation	Illustrations, thumbnails, character art	Imagen 4 (Google), Flux 1.1 Pro , Midjourney v7	DALL-E 3, Stable Diffusion XL
3D / simulation assets	Interactive or 3D visuals (where the topic warrants, e.g. Science)	Blender (scriptable), Spline , PhET simulations (embed), Three.js	Unity
Voice cloning — English	Embibe-branded narrator voice	ElevenLabs v3	Google Chirp 3 HD, OpenAI TTS
Voice cloning — Hindi / Indic	Natural Hindi / regional narration	Sarvam AI TTS (Indic-native), ElevenLabs v3	AI4Bharat TTS, Google Chirp 3 HD

Stage	Purpose / outcome	Proposed tools & models	Alternates
Text-to-video / avatar	Assemble narrated video with visuals	HeyGen or Synthesia (avatar-led, best for explainers); Google Veo 3 / Pika, D-ID Runway Gen-4 / OpenAI Sora 2 (cinematic b-roll)	
Video composition	Stitch slides, voice, diagrams, captions into the final 1080p video	Remotion (React-based, programmatic, repeatable) or FFmpeg	CapCut API, Descript
Captions	Burned-in or sidecar captions	Whisper v3 Turbo for forced alignment + manual QA	AssemblyAI
Key moments (jump-to)	3–6 logical segments with titles, start/end timestamps	Gemini 2.5 Pro over timestamped transcript (pattern already used in Superr video enrichment)	—
Provenance / watermark	Tamper-evident record of what was AI-generated and how	C2PA signed manifest; Google SynthID watermarking for images and audio	—

6.3 Shared / orchestration layer

Stage	Purpose / outcome	Proposed tools & models	Alternates
Pipeline orchestration	Chain stages, retries, branching, human-in-loop gates	LangGraph or Temporal (durable, observable, production-ready)	n8n, Prefect
Prompt / eval tooling	Track prompt versions, run A/B evals, regression tests	PromptFoo + Langfuse (tracing)	Helicone, Arize Phoenix
LLM-as-judge (evaluation)	Score every output against the scorecard, flag issues for human reviewer	Gemini 2.5 Pro (cheap, long context, multimodal — watches the video) + Ragas / DeepEval as the eval harness	Claude Sonnet 4.6 as a secondary judge for cross-check
Embeddings	Tag outputs for Embibe search / Superr Chat retrieval	gemini-embedding-001 (multilingual, strong Indic); BGE-M3 (self-hosted option)	OpenAI text-embedding-3-large
Vector store	Fast semantic retrieval for the curated / generated corpus	pgvector (already in Embibe stack)	Qdrant, Pinecone
Storage	Video, PDF, transcript, GCS artifacts	GCS / S3 (existing Embibe infra)	—
Observability	Cost, latency, error rates, hallucination rate per stage	Langfuse for LLM spans; Grafana for infra dashboards	Helicone

Note: All model choices reflect best-available tooling as of Jan 2026. Phase 1 includes a bake-off step where the team runs trial outputs through two candidate models per stage (where alternates are listed) and the Content Quality reviewer + PM pick the winner before freezing the stack.

7. How We Evaluate Output

7.1 Scoring steps

Step 1 — Automated scoring (LLM-as-judge) An LLM scores every topic against the scorecard below and flags factual errors, LO drift, and anything scoring low for human review.

Step 2 — Human review (Content Quality reviewer) The Content Quality reviewer (human-in-loop) spot-checks every topic, validates the LLM scores, and makes the final pass/fail call.

7.2 Scorecard

Each criterion is scored 0–5 and combined via the weights below.

Criterion	Wt.	Check
Factual accuracy	25%	Every claim correct. One error = hard stop.
Curriculum alignment	20%	Covers every input LO, stays on-syllabus.
Pedagogical quality	15%	Opening, examples, check-for-understanding, recap.
Age appropriateness	10%	Vocabulary and examples fit the grade.
Engagement	10%	A student in that grade would stay through it.
Production quality	10%	Clean audio, captions, visuals, working links.
Language fidelity	5%	Natural phrasing, correct grammar (Hindi/regional only).
Safety & bias	5%	No religious/political/gender/caste/commercial bias.

How each is checked: LLM-as-judge scores every criterion against the NCERT reference (claims, LO coverage, structure, readability, pacing, audio/caption/link checks, bias classifier). The Content Quality reviewer validates LLM scores, spot-checks facts and clips, and makes the final call. Language fidelity uses a native-speaker reviewer; skipped for English-only (weight redistributed).

7.3 Publish-ready threshold & decision rules

Publish-ready threshold: weighted score $\geq 3.8 / 5.0$ (76%), no single criterion below $3.0 / 5.0$ (60%), zero factual errors.

What counts as a factual error: - Incorrect dates, names, places, definitions, or formulae - Misstated cause-effect, sequence, or relationships (e.g., wrong chronology, wrong branch of government) - Hallucinated sources, quotes, or statistics - Misalignment with NCERT/prescribed textbook content for the same topic - Any claim that contradicts the reference material provided as input

Decision: - $\geq 3.8 / 5.0$ (76%), zero factual errors $\rightarrow$ green-light scale-up for that approach $\times$ gap type - Below $3.8 / 5.0$ (76%) $\rightarrow$ iterate and re-run, or drop the approach for that gap - Any factual error $\rightarrow$ hard stop, do not publish

8. Resources Needed

Lean POC team. 4–5 people total.

Role	Count	Scope / Skills
POC lead (PM)	1	End-to-end owner, weekly reviews, decision memo

Role	Count	Scope / Skills
Engineer	2	One runs the curation pipeline + NCERT PDF ingestion; one runs the generation stack (scripting, video, voice). Skills: LLM orchestration, video/voice tooling, Python
Human-in-loop		
— Content	1 or 2	Validates LLM scorecard ratings, does final pass/fail review, flags factual errors
Quality		

Tools & platforms - Curation stack: existing Superr backend (YouTube scraper, curriculum graph, tagging) + lightweight NCERT PDF ingestion module - **Generation stack:** LLM for scripting, text-to-video, voice cloning (English + Hindi), image/diagram generation - **Transcription:** Whisper for English; IndicWhisper or equivalent for Hindi; regional language models (Tamil, Bengali, Telugu, Marathi, etc.) to be added as scope expands - **Evaluation:** LLM-as-judge pipeline running the scorecard; shared scoring sheet for human-in-loop validation - **Storage:** existing Embibe content DB + object storage (GCS / S3 for video, PDF, and transcript artifacts)

9. Execution Plan

Phase 1 — Setup

Get the inputs, tools, and scoring prompt ready so the team can actually start running experiments.

1. Lock topic IDs, learning objectives, and reference NCERT PDFs for all 4 experiments (TOC)
2. Finalize the YouTube source channel list for curation
3. Build the NCERT PDF reader module so the curation pipeline can ingest textbook pages
4. Get the generation tools ready — pick and set up the LLM for scripting, the text-to-video tool, and the voice-cloning tool
5. Set up the automated scoring — write the scorecard as an LLM prompt and run it on a sample to confirm it scores sensibly

Phase 2 — Calibration

Run two trial topics, compare LLM scores against the reviewer's scores, and tune the LLM prompt until the two agree.

6. Run 1 topic through the curation approach as a trial
7. Run 1 topic through the generation approach as a trial
8. Content Quality reviewer watches/reads both trial outputs end-to-end and fills a simple scoring sheet — one row per topic, one column per criterion, short notes on anything that looks off
9. Engineer runs the LLM on the same two outputs and drops its scores into the same sheet alongside the reviewer's
10. Reviewer, PM, and engineer walk through the sheet together — flag every place the LLM missed a factual error the reviewer caught, over-scored a criterion, or used signals the reviewer disagrees with
11. Update the LLM prompt for the flagged criteria (e.g., “also check for NCERT chronology”, “weigh the opening more heavily for engagement”)
12. Re-run the LLM on the same two outputs and walk through again, until the reviewer is comfortable the LLM flags what it should
13. Freeze the scoring sheet and the LLM prompt before Phase 3 begins

Phase 3 — Run

Produce content for all four experiments using the approach assigned to each.

14. Run Experiments 1 and 2 through the curation approach
15. Run Experiments 3 and 4 through both curation and generation

Phase 4 — Score

Score every topic, LLM first and then human reviewer, to get one consistent quality read per output.

16. LLM scores every topic and prepares the list for Content Reviewer
17. Content Quality reviewer (human-in-loop) validates, flags factual errors, and makes the final pass/fail call

Phase 5 — Decide

Look at the scores, pick which approach scales into which gap, and take the go/no-go to leadership.

18. Per-experiment pass/fail against the 3.8 / 5.0 (76%) threshold
19. Recommendation on which approach scales into which gap type, with a phased roadmap
20. Leadership go/no-go